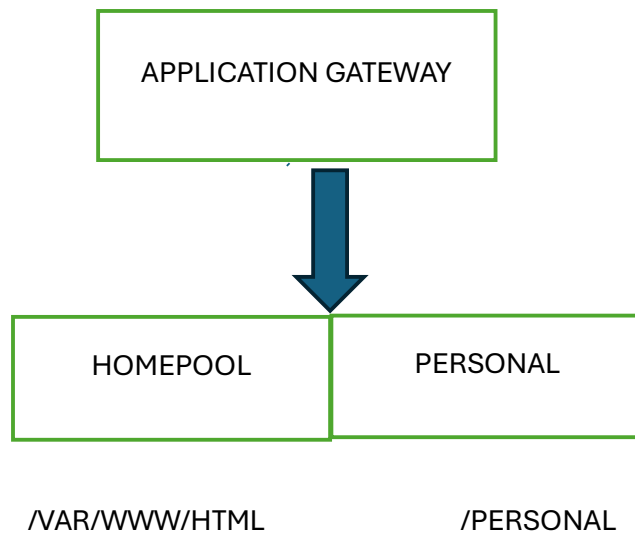


APPLICATION GATEWAY

1. Application gateway is nothing but a Regional Load Balancer.
2. It Routes Path/URL based routing.
3. It works on OSI layer 7 called application layer.
4. It supports both HTTP and HTTPS based applications.



VNET-1

SUBNET-1: VM-1 + VM2

SUBNE-2: APPPLICATION GATEWAY

1. connect VM-1 by using ssh and browse on browser by using public IP.
2. Next for second machine we have to add path called /personal. When we browse second machine on browser by using public IP with /personal the result needs to get.

The screenshot shows the Microsoft Azure portal interface. The left sidebar displays the 'Virtual machines' section. The main pane shows the details for 'Sudheer-VM01'. The 'Overview' tab is selected, showing the VM's status as 'Running'. The 'Essentials' section provides a summary of the VM's configuration, including its resource group, location, subscription, and network settings. The VM is connected to the internet, and its public IP address is 20.118.130.139.

The screenshot shows a web browser window displaying the 'Welcome to nginx!' page. The address bar shows the URL 'http://20.118.130.139'. The page content includes a message about the nginx web server being successfully installed and working, along with links to online documentation and support.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

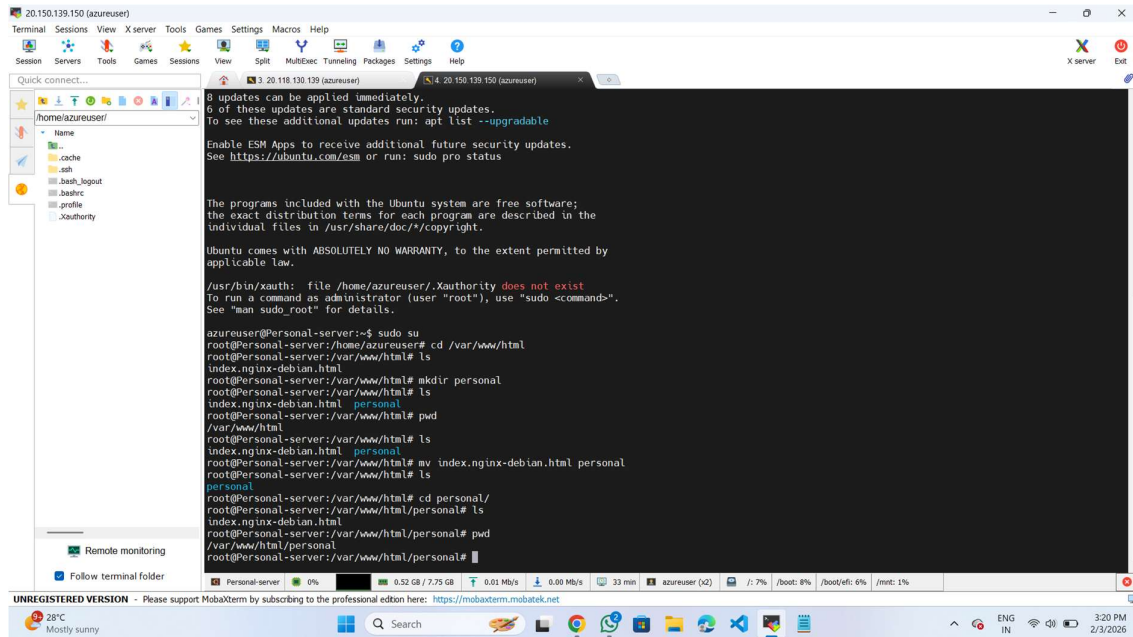
Thank you for using nginx.

The screenshot shows the Microsoft Azure portal interface. The left sidebar displays the 'Virtual machines' section. The main pane shows the details for 'Personal-server'. The 'Overview' tab is selected, showing the VM's status as 'Running'. The 'Essentials' section provides a summary of the VM's configuration, including its resource group, location, subscription, and network settings. The VM is connected to the internet, and its public IP address is 20.150.139.150.

Not secure http://20.150.139.150/personal

404 Not Found

nginx/1.24.0 (Ubuntu)



```
8 updates can be applied immediately.
6 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

/usr/bin/xauth: file /home/azureuser/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@Personal-server:~$ sudo su
root@Personal-server:/home/azureuser# cd /var/www/html
root@Personal-server:/var/www/html# ls
index.nginx-debian.html
root@Personal-server:/var/www/html# mkdir personal
root@Personal-server:/var/www/html# ls
index.nginx-debian.html  personal
root@Personal-server:/var/www/html# pwd
/var/www/html
root@Personal-server:/var/www/html# ls
index.nginx-debian.html  personal
root@Personal-server:/var/www/html# mv index.nginx-debian.html personal
root@Personal-server:/var/www/html# ls
personal
root@Personal-server:/var/www/html# cd personal/
root@Personal-server:/var/www/html/personal# ls
index.nginx-debian.html
root@Personal-server:/var/www/html/personal# pwd
/var/www/html/personal
root@Personal-server:/var/www/html/personal#
```

Not secure http://20.150.152.173/personal/

Welcome to My Application Gateway!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

1. When I browse with second vm public IP/personal the result has not get. Next I have added /personal and browse with public ip/personal the result has shown in the above figure.

- Next I have created a application gateway as shown in figure below.

The first screenshot shows the 'Create application gateway' form in the Azure portal. The form is titled 'Create application gateway' and includes the following fields and options:

- Instance details:**
 - Application gateway name: APP-LB01
 - Region: West US 3
 - Tier: Standard V2
 - Enable autoscaling: ☐ Yes ☒ No
 - Instance count: 1
 - IP address type: ☒ IPv4 only ☐ Dual stack (IPv4 & IPv6)
 - HTTP2: ☒ Disabled ☐ Enabled
 - FIPS (Federal Information Processing Standard) mode 140-2: ☒ Disabled ☐ Enabled
- Configure virtual network:**
 - Virtual network: vnet-westus3
 - Subnet: SUBNET-02 (172.18.1.0/24)

The second screenshot shows the 'Overview' page for the created Application Gateway 'APP-LB01'. The page includes a sidebar with navigation options like 'Activity log', 'Access control (IAM)', 'Tags', 'Diagnose and solve problems', 'Resource visualizer', 'Settings', 'Monitoring', 'Automation', and 'Help'. The main content area displays the following details:

- Essentials:**
 - Resource group (move): Sudheer-RG
 - Location: West US 3 (Zone 1, 2, 3)
 - Subscription (move): Azure subscription 1
 - Subscription ID: a3107e0d-376f-4650-b383-aa7b2b3c0c9a
 - Tags (edit): Add tags
- Virtual network/subnet:** vnet-westus3/SUBNET-02
- Frontend public IP address:** 20.150.152.173 (apilbpbip)
- Frontend private IP address:** -
- Tier:** Standard V2
- Availability zone:** 1, 2, 3

Below the details, there is a green checkmark icon and the text: 'Your resource is following Light best practices. Azure Advisor provides personalized recommendations to reduce costs, increase security, optimize performance and reliability and achieve operational excellence. View recommendations for all your resources in Azure Advisor.'

1.After creating application gateway. In application gateway I have configured some steps called backend pools, backend steps, listeners, frontend ip, rules,Health probes etc. all this images are mentioned below

Microsoft Azure | APP-LB01 | Backend pools

Search resources, services, and docs (G+V)

Home > Load balancing and content delivery | Application gateways > APP-LB01

Overview

Load balancing

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

Showing 1 - 1 of 1. Display 20

APP-LB01 | Backend pools

Search backend pools

Name	Rules associated	Targets
ig-backend-pool	1	0
Home-pool	1	1
Personal-pool	1	1

28°C Mostly sunny

Microsoft Azure | APP-LB01 | Backend settings

Search resources, services, and docs (G+V)

Home > Load balancing and content delivery | Application gateways > APP-LB01

Overview

Load balancing

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

Showing 1 - 1 of 1. Display 20

APP-LB01 | Backend settings

Search Backend settings

Name	Port	Protocol	Cookie based affin...	Custom
ig-settings	808	Http	Disabled	-
home-setting	80	Http	Disabled	Home-hj
personal-setting	80	Http	Disabled	personal

28°C Mostly sunny

Microsoft Azure | APP-LB01 | Frontend IP configurations

Search resources, services, and docs (G+V)

Home > Load balancing and content delivery | Application gateways > APP-LB01

Overview

Load balancing

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

Showing 1 - 1 of 1. Display 20

APP-LB01 | Frontend IP configurations

Search frontend IP configurations

Type	Status	Name	IP address	Associated
Public	Configured	appGwPublicFrontend...	20.150.152.173 (appb...	ig-listener...
Private	Not configured	-	-	-

28°C Mostly sunny

Microsoft Azure | APP-LB01 | Listeners

Listeners

SSL Policy

Selected SSL Policy: Default (change)

Policy name: AppGatewayPolicy2020101

Min protocol version: TLSv1.2

Cipher suites: TLS_AES_128_GCM_SHA256

Name	Port	Protocol	Frontend IP	Associate
app-listener	80	HTTP	Public IPv4	Rule
ig-listener	806	HTTP	Public IPv4	ig-rule

Microsoft Azure | APP-LB01 | Health probes

Health probes

Name	Protocol	BackendSettings	Host or SNI value	Path
personal-hp	HTTP	> 1	20.150.139.150	/personal
Home-hp	HTTP	> 1	20.118.130.139	/

Microsoft Azure | APP-LB01 | Rules

Rules

Name	Type	Listener	Priority
Rule	Path-based	app-listener	100
ig-rule	Basic	ig-listener	800

This is the final answer I got when I browse with application gateway id.

⚠ Not secure http://20.150.152.173/personal/

Welcome to My Application Gateway!

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Commercial support is available at nginx.com.

Thank you for using nginx.